

Unit II: Linear Algebra

Question Bank

- Check whether the following are linear transformations/operators where v denotes the vector (v_1, v_2, v_3, v_4) belonging to $\mathbb{R}^4$.
 - $T(v) = 3v + S(v)$ where S is a linear operator on $\mathbb{R}^4$
 - $T(v) = (v_1+v_2, v_3+2, v_4)$
 - $T(v) = v \cdot v_0$ (inner product) where v_0 is a fixed vector in $\mathbb{R}^4$.
 - $T(v) = S_1(S_2(v))$ where $S_2 : \mathbb{R}^4 \rightarrow V$ and $S_1 : V \rightarrow W$ are linear transformations, with V and W , vector spaces.
- Let $T(x_1, x_2, x_3, x_4) = (2x_1+x_3+x_4, x_1+2x_2+x_4, x_1+x_2+2x_3, x_2+x_3+2x_4)$ w.r.t. the basis $B_1 = \{(1,0,0,0), (2,1,0,0), (1,2,1,0), (1,1,2,1)\}$ for domain and $B_2 = \{(0,0,0,1), (0,0,1,2), (0,1,2,1), (1,2,1,1)\}$ for co-domain.
- Find the matrix of $T(x, y, z) = (x, x + y, y + z)$ w.r.t., the standard ordered basis for the domain and the basis $\{(1,2,1), (2,1,2), (3,3,4)\}$ for the co-domain.
- Let S be rotation by 90 degrees and T the reflection across the line $y = x$.
 - For $v = (2,5)$, calculate $TS(v)$ and $ST(v)$.
 - Can you find a non-zero vector w such that $TS(w) = ST(w)$?
- Let S be the reflection across the y -axis and T the projection on the x -axis.
 - Is $ST = TS$?
 - What if S is instead the reflection across the line $y = -x$?
- Find the range, rank, kernel, nullity of T in the following cases
 - T is the linear transformation given in Question 2 above
 - $T(x_1, x_2, x_3, x_4) = (x_1+x_4, x_1+x_2, x_2+x_3, x_3+x_4)$
- Find a set of vectors which span $S^\perp$ for the following cases of S .
 - $S = \{(1,4,-2)\}$
 - $S = \{(1,4,3,5), (-6,2,1,4)\}$
 - $S = \{(3,1,4), (1,2,3), (5,2,3)\}$

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- 8 Find the plane through the origin in $\mathbb{R}^3$ that is orthogonal to the intersection of the planes $x-2y=0$ and $y-3z=0$.
- 9 Find the 2-d plane through the origin in $\mathbb{R}^4$ that is orthogonal to the plane by the system $x - y + z + t = 0$, $3x + 5z - t = 0$.
- 10 Find the matrix for the projection onto
 - a the vector $(-3,1,-2)$.
 - b the subspace spanned by $(5,2,7)$ and $(6,4,6)$.
 - c the x_2x_4 -plane in $\mathbb{R}^4$ whose vectors have coordinates (x_1, x_2, x_3, x_4)
 - d the vector $(9,8,7,6)$.
- 11 Find the projection matrix:
 - a If B is the projection matrix for projection onto a subspace V of $\mathbb{R}^n$, what is the matrix for the projection onto $V^\perp$?
 - b Calculate the projection matrix for projection onto the left-null space of a matrix A , using the projection matrix for the same onto $C(A)$.
 - c What are the matrices for the projections onto the row space and the null space of A ?
- 12 Find the straight line that best fits the following data:

x	0	3	5	8	10
y	2	4	7	12	17

- a Find the straight line that best fits the data given above.
- b Can you find the quadratic $y = ax^2 + bx + c$ that best fits the above data? (HINT: Write matrix A with all 1's in the first column, x values in the second and x^2 values in the third).
- c What are the dimensions of the matrix A in the above hint, if we look for a model of the form $y = p(x)$ where p is a polynomial of degree k ?